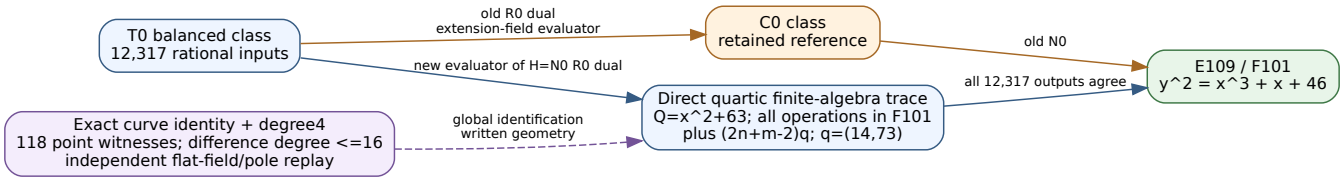
A direct F101 quartic quotient evaluator

2026-11-12 · fixed T0→E109 · scientific standing unsigned

The existing quotient H=N0 R0 dual now has an explicit direct evaluator overF101. It agrees with the retained route on all12,317 balanced T0 classes,all negations and predetermined paired sums,including the actual closed poles,repeated support and every infinity case,PCC585f35. The old route remains the reference. This is the first elliptic component; the rest of the Jacobian chain retains its existing evaluation paths. [pcc:unverified]



Exact coefficients and complete finite identity

Source F=[31,75,74,54,71,55,1];target $Y^2=X^3+X+46$,overF101. All lists are low to high. The cover is(X,Y)=(N/Q^2,y M/Q^3),where

N	14,0,88,97,14
Q	63,0,1
M	41,40,69,73
infinity+ image q	(14,73)

The exact polynomial identity is F $M^2=N^3+N\,Q^4+46Q^6$. Existing field/polynomial engines check every coefficient,gcd,smoothness and nonzero derivative. The reduced rational x-map has degree4;the complete curve map therefore has degree4. The infinities map to±q. Each of the four simple geometric pole branches maps to the elliptic origin. Native constructionad6883 and separate flat-field replay5ab2bd retain the source and all118 finite rational witness points. [pcc:unverified]

Every balanced class and exceptional route

The input finite degree is m and plus-infinity count n. The finite trace needs the exact correction(2n+m-2)q. Degree0 plus0/1/2 yields-2q/O/+2q. Finite quadratic support uses the existing polynomial inverse modulo u, then the actual elliptic secant or its coincident-image limit. NonunitQ uses gcd(u,Q^2),including multiplicity,so poles are never inverted. [pcc:unverified]

Degree0	Linear support	Closed-pole support	Generic quadratic
3	236	2	11,842
Same image: double	Opposite images	Repeated supports	Total classes
126	108	118 (overlapping count)	12,317

A global certificate rather than an interpolation assumption

The old quotient has HH†=[4],from the accepted Richelot dual norm and elliptic norm/pullback identities. Its Abel–Jacobi curve map has degree4. Centre it by q=H(infinity+-infinity-)/2,using the actual odd elliptic order109 and scalar55. The centred map is odd under the hyperelliptic involution. Irreducibility of the source sextic and the degree bound force its finite denominator to be a square;finite infinity values give the displayed degree bounds. [pcc:unverified]

For two degree4 curve maps f,g to E,Milne AVv2 ChapterI Corollary5.3, applied to f,g,-g and the symmetric line bundle O_E(O),gives

deg(f+g)+deg(f-g)=2deg f+2deg g=16.

A nonconstant curve map has at most its degree many distinct points in one fibre. Hence17 distinct agreements force f-g to be constant,and one agreement fixes that constant to zero. The frozen certificate has118 distinct agreements. This uses a proved degree bound,not density of rational samples. The old H expected-value provenance is retained as an explicit dependency. QUARTIC_COVER_PROOF.md/PCC6afa9c contains the full written argument and complete balanced finite-algebra formulas. [pcc:unverified]

Independent replay and a discriminating negative control

A separate checker imports existing flatFpR/polynomial operations and no producer Balanced,Richelot or interpolation code. It checks the exact identity,source irreducibility,degree/separability,all118 complete finite rational points,both infinity limits and all four geometric poles. Mutated numerator and codomain controls reject. Negating M still gives the exact curve equation but disagrees on118 witness points. This shows why the equation alone cannot identify the intended quotient. Checker04 passes2tests/strictClippy5ab2bd;the complete standalone10 passes15tests with prior class/kernel controls,c3a4b7. [pcc:unverified]

Existing code, integration and continuation

All field,polynomial,elliptic,Richelot and class arithmetic is reused. The private existing Sutherland Lagrange body is extracted byte-exact with visibility-only custody. Compiler01 used the wrong SevenEdge field; its source/error7137fa remains. No degree bound or test was weakened. First-component actual consumer integration05 passes2targeted tests/ strictClippy86b79a. Complete changed guard07 passes25native/85reference unit+five integrations,one old ignored benchmark,all source guards/fmt/ strictClippy301fbe. Complete108004-byte raw standard-XZ12816bytes has raw/archive hashes and exact roundtrip;post-run source05 hashes match. Parent1501 exact5ceb focused43157257094 succeeds25/83+five,fullf74e9e. Own final-headCI and bottom-up integration remain separate. [pcc:unverified]

Formalization remains partial:the existing134 written Lean declarations are unchanged;actual Picard,polarization,curve degree and line-bundle geometry are written dependencies here. The second T0→E113 norm has separate full12317 acceptance2fcf0b and joint H0 forward comparison 19b3bd;all four T0/T1 pullbacks pass5143f5. Broader isolated evaluation work is separate from this first-component consumer change. Matched cold costs await S3/livebudget admission; this report supplies no measured cost ratio. [pcc:unverified]

Primary sources read:Milne AVv2 Corollary5.3(the arbitrary-map cube identity);Galbraith corrected Chapter8 Lemma8.1.9,Theorem8.2.12/ Corollary8.2.13 and Theorem8.3.8. Exact source links/hash and old global dependencies are in6afa9c. KB/ecdb unavailable;atlas vault locked. [pcc:unverified]